



In vivo toxicity evaluation of aqueous acetone extract of *Terminalia bellirica* (Gaertn.) Roxb. fruit



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ABSTRACT

Terminalia bellirica (Gaertn.) Roxb. (Family: Combretaceae), known as Bhibhitaki in Sanskrit and locally known as Behera in India is one of the oldest medicinal plants which has widely been used in the traditional system of medicine, especially in Ayurveda for centuries. The dried fruit of *Terminalia bellirica* is used for treating various ailments. Aqueous acetone extract of *Terminalia bellirica* (Gaertn.) Roxb fruits (AATB), showed antioxidant potential in our screening study is selected for the present *in vivo* toxicity evaluation. Acute administration of AATB was done in female Wistar Albino rats as a single dose up to 2000 mg/kg body weight. At the end of the study, Blood was collected for biochemical and hematological analyses, while histological examinations were performed on liver and kidney. There was no alteration in the behavioral pattern, food and water intake in the treated animals. The relative organ weight, biochemical parameters, hematological parameters and histopathological analysis were also found normal. All the parameters of the toxicity evaluation were found to be normal and the data suggests aqueous acetone extract of *Terminalia bellirica* fruit is safe, to be used as a traditional herbal formulation for its antioxidant potential and other health benefits.

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1. Introduction

Traditional systems of medicine like Ayurveda, Unani, Siddha play a vital role in mitigating various ailments of humankind especially those who inhabit rural areas. The demand for plant-based phytotherapeutics is driven by their effectiveness, fewer or no side effects and above all cost effective. In India, out of 45,000 plant species identified with medicinal properties, around 20 thousand medicinal plants are of significant medicinal value and therapeutic potential (Deb et al., 2016). Plants used in the traditional system of medicine are a rich source of phytochemicals and have provided many valuable therapeutics which are now used in modern medicine such as digitoxin, opium, morphine, taxol, vincristine and the like (Motamarri et al., 2012). Phytoconstituents present in medicinal plants act synergistically to produce larger physiological benefits in human beings than the individual effects of each component. India has a rich tradition of the plant-based medicinal system of treating various diseases. Ayurveda is one of the oldest healthcare systems that evolved in the Indian

subcontinent.

Terminalia bellirica (Gaertn.) Roxb. (Combretaceae), known as Bhibhitaki in Sanskrit and locally called Behera in India is a large deciduous tree found throughout India except for the dry western region up to an elevation of 900 m (Hazra et al., 2010). The fruits, apart from being an integral part in Triphala, are widely used in various traditional therapeutic formulations in the indigenous system of medicine either alone or in combination with other plant-based drugs (Meena et al., 2009). The fruits have laxative, astringent, anthelmintic and antipyretic properties and are used in Ayurveda against various disorders like hepatitis, bronchitis, asthma, dyspepsia, piles, diarrhea, coughs, eye diseases (Deb et al., 2016). The fruits elicit various pharmacological properties like antioxidant, antidiabetic, analgesic, antidiarrhoeal in its different extracts (Deb et al., 2016).

Herbal antioxidants have vital role in the prevention and management of various oxidative stress related proliferative and degenerative disorders. Herbal remedies, even though they have become more popular in the perception of harmless owing to their natural origin, they too possess adverse effects (Ernst, 1998). Hence it is also essential for the herbal formulations to have scientific toxicological safety assessment before being developed into new herbal therapeutic medicines. In our previous study, aqueous

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acetone extract of *Terminalia bellirica* fruit exhibited maximum antioxidant activity among the screened extracts (Latha et al., 2016). Therefore, the present study was conducted to evaluate the *in vivo* acute toxicity of aqueous acetone extract in experimental rodent models.

2. Materials and methods

2.1. Plant material

Authenticated fruits collected during April–May 2015 from wild plants of *Terminalia bellirica* were purchased from Kerala Forest Research Institute, Peechi, India. The specimen sample was deposited at the institute with voucher number SBSBRL 23. After drying in shade for few days, the fruits were de-seeded, finely powdered and stored in an airtight container until used for extraction.

2.2. Preparation of extracts

The fruit powder (100 g) was defatted with petroleum ether and then extracted with 70% aqueous acetone (300 ml) in a mechanical shaker for 72 h. The extracts were filtered through Whatman No.1 filter paper, concentrated to dryness in a rotary evaporator, lyophilized and then stored at 4 °C until use. The percentage of yield was found to be 6.23%.

2.3. Experimental animals

Healthy female Wistar albino rats weighing between 120 and 150 g were used for the study. The animals were procured from a licensed breeder (Nagarjuna Herbal Concentrates Private Limited, Thodupuzha). The animals were housed in polypropylene cages, maintained on a 12-h light/dark cycle and provided with standard pellet and water *ad libitum*. The animals were acclimatized to the laboratory conditions for one week before starting the experiment. The study protocol was approved by the Institutional Animal Ethics Committee at School of Biosciences, Mahatma Gandhi University, Kottayam in accordance with the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA) guidelines (Reg No B23032014-02) and all experiments were carried humanely.

2.4. Acute toxicity study

The acute toxicity of the aqueous acetone extract of *Terminalia bellirica* (Gaertn.) Roxb. fruit was evaluated in rodent models according to the Organisation for Economic Co-operation and Development (OECD) guideline 425 (“[Oecd guidelines for the testing of chemicals, 2001](#)”).

The animals were divided into six groups (6 rats in each) where Group I served as normal control (untreated). Group II, III, IV and V were orally administered with single dose of 55, 175, 550 and 2000 mg/kg body weight of the extract. The animals were fasted overnight prior to dosing but allowed free access to water. After taking the fasting body weight of each animal, a single dose of the extract was administered via oral gavages. The rats in the control group received distilled water. The food was withheld for 3 h after administering the extracts. The animals were closely monitored and observed for any signs of toxicity for the entire study period.

Animals were sacrificed after overnight fasting on the 15th day and blood was collected. The organs were cleaned with normal saline, weighed and preserved in 10% formalin for further analysis.

2.5. Clinical signs

The animals were closely monitored and observed for any signs of toxicity such as behavioral changes, physical appearances and mortality immediately after dosing, at 4 h and at 24 h intervals and then twice daily for 14 days. The parameters included changes in the skin, fur, eyes, mucous membranes and various behavioral patterns of the treated animals. Treated animals were observed for any tremors, convulsions, salivation, diarrhea, lethargy, sleep, coma and mortality.

2.6. Food and water consumption

The amount of food and water intake of treated animal was observed for 14 days and the data was compared with normal animals.

2.7. Body weight

The weight of individual animals was recorded before administering the dose and is taken as the weight of the animal at day1. The weight of the animals was recorded later on 7th and 14th day of the experiment. Weight gain of the treated animals was compared with untreated normal rats.

2.8. Relative organ weight

The vital organs, liver, pancreas, kidney and heart were removed soon after sacrificing the animals on the 15th day, cleaned with saline and their wet weight was determined. Relative organ weight was calculated as (weight of organ/body weight of rat on day of sacrifice) × 100% (Mohamed et al., 2011).

2.9. Hematological parameters

Blood samples collected in EDTA tubes were used for hematological analyses including Total count (TC), total white blood cell (WBC), red blood cell count (RBC), platelet count, hemoglobin (Hb) and erythrocyte sedimentation rate (ESR).

2.10. Liver tests

The blood collected in clot stimulating tubes were kept at room temperature for 30 min to separate the serum. The serum was transferred to fresh tubes and stored at –20 °C until use. The serum samples were used to determine biochemical markers of liver function such as aspartate transaminase (AST), alanine transaminase (ALT), gamma glutamate transaminase (GGT), alkaline phosphatase (ALP), lactate dehydrogenase (LDH), total and conjugated bilirubin, total protein, albumin, globulin and A/G ratio using standard diagnostic kits (Span Diagnostics Limited kits, Surat, India).

2.11. Renal function tests

The serum samples separated from the blood samples of the sacrificed animals were used to determine renal function assays such as creatinine, blood urea and uric acid. These tests were performed using standard diagnostic kits (Span Diagnostics Limited kits, Surat, India).

2.12. Antioxidant status

Liver tissue was homogenized in phosphate buffer (100 mM) containing EDTA (1 mM) to make 10% tissue solution. The solution

Table 1

Clinical signs in acute oral toxicity study of AATB.

Clinical signs	Normal	AATB-55	AATB-175	AATB-550	AATB-2000
General behaviour	Normal	Normal	Normal	Normal	Normal
Sleep	Normal	Normal	Normal	Normal	Normal
Tremors	NIL	NIL	NIL	NIL	NIL
Skin & Fur	Normal	Normal	Normal	Normal	Normal
Diarrhea	NIL	NIL	NIL	NIL	NIL
Mortality	NIL	NIL	NIL	NIL	NIL
Lethargy	NIL	NIL	NIL	NIL	NIL

was centrifuged at 12,000 g for 30 min at 4 °C and the supernatant obtained was used for the analysis of reduced glutathione (GSH) (Ellman, 1959), superoxide dismutase (SOD) (Kakkar et al., 1984), catalase (Aebi, 1984) and lipid peroxidation (Ohkawa et al., 1979) to determine the antioxidant status (Khan et al., 2013).

2.13. Histopathology

Small pieces of liver and kidney tissues, harvested from the animals used in the study, were fixed in 10% buffered formalin and further processed using automated tissue processor. The paraffin embedded tissue blocks were made 5 µm thick sections using a rotary microtome and dried overnight in an oven at 37 °C. The sections were later stained with hematoxylin and eosin (H&E) and examined under the microscope to study toxicity associated histological changes in the liver and kidney tissues.

2.14. LD₅₀ value

LD₅₀ value of the extract (AATB) was calculated using the Acute Oral Toxicity (Guideline 425) Statistical Program (aot425).

2.15. Data analysis

The results were expressed as mean ± SD where each value represents a minimum of six rats (n = 6). The data for relative organ weights, hematology and serum biochemistry were tested for one-way analysis of variance (ANOVA) followed by Dunnett's multiple comparison tests in which the results were compared with that of normal rats. The results were considered statistically significant at $p \leq 0.05$ level. GraphPad Prism® version 5.03 for Windows (GraphPad Software, USA) was used for all statistical analysis.

3. Results

3.1. Behavioral patterns and clinical signs

There were no significant changes observed in the clinical signs as well as behavioral patterns of the treated animals (Table 1).

3.2. Food and water consumption

The food and water consumption was measured throughout the study period. It has been observed that there is no significant difference ($p > 0.05$) between the normal group and the treated groups (Fig. 1).

3.3. Body weight measurement

The body weight of rats was measured on the first day, 7th day and 14th day of the acute toxicity study and no significant ($p > 0.05$) difference in the body weight was observed between control and

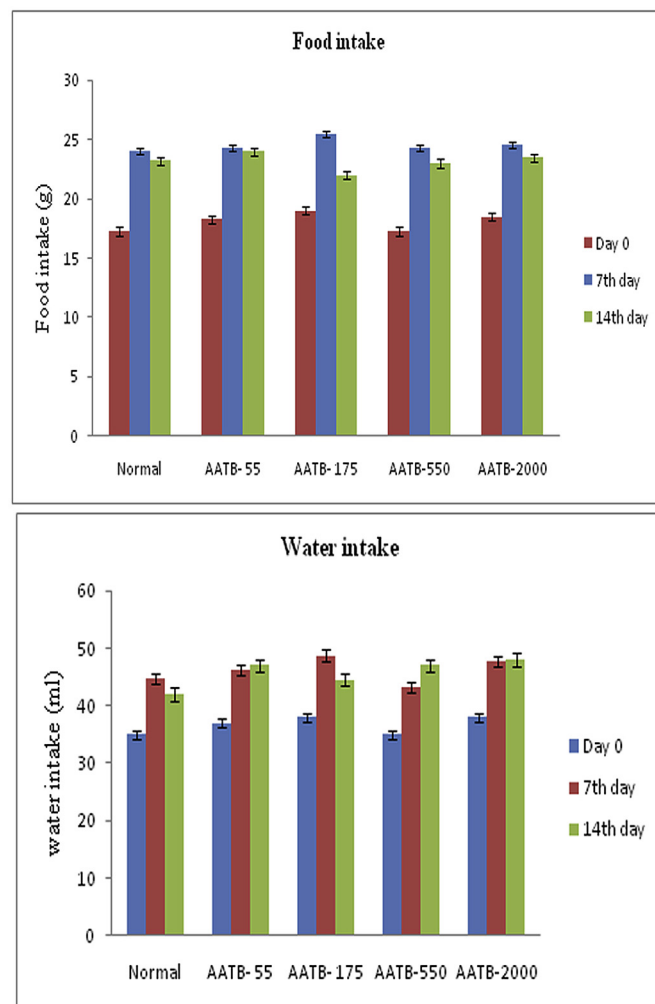


Fig. 1. Consumption of food and water by the control and AATB treated rats during the acute toxicity (for 14 days) study. Values are expressed as mean ± standard deviation (n = 6 for each group).

AATB treated rats (Fig. 2).

3.4. Relative organ weight

There was no significant change ($p > 0.05$) in the weight of liver, kidney, heart and pancreas of the treated and un-treated rats (Table 2). The relative organ weight of the rats of treatment, as well as normal groups, also remained insignificant.

3.5. Hematological parameters

The hematological parameters used in the toxicity study were Total count (TC), total white blood cell (WBC), red blood cell count (RBC), platelet count, hemoglobin (Hb) and erythrocyte sedimentation rate (ESR). The blood parameters remained in the normal range with no significant difference between the rats of treatment and nontreatment groups (Table 3).

3.6. Liver tests

The results of liver function markers (AST, ALT, GGT etc) used in the acute toxicity study (Tables 4 and 5) showed no significant change ($p > 0.05$) between the normal and AATB administered rats.

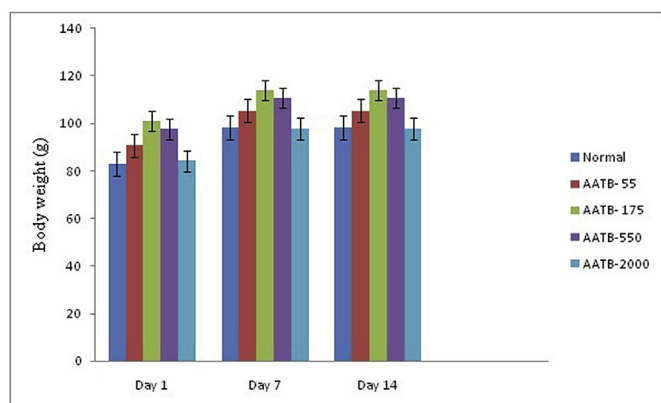


Fig. 2. Mean body weight of rats receiving AATB extract (Dose in mg/kg body weight) for 14 days. Values are expressed as mean $\pm$ standard deviation ($n = 6$ for each group).

3.7. Renal function tests

The Table 6 represents the results of renal function tests used to assess the kidney function status during the acute toxicity study. The AATB treated rats even in the highest concentration showed no significant change in the levels of renal toxicity markers, creatinine, urea and uric acid when compared with the normal rats.

3.8. Antioxidant status

The result of antioxidant status evaluation of AATB treated animals (Table 7) showed no signs of oxidative stress. The markers of tissue oxidative stress of the present study include SOD, catalase, GSH and MDA which all showed comparable levels with normal rats.

3.9. Histopathology

Histopathological analysis of the vital organs, liver and kidney of the AATB treated animals showed no altered histopathology and signs of toxicity. The liver showed normal architecture with a clear lumen of central vein and without any lesion or necrosis (Fig. 3). There was no renal injury evident in the histopathological evaluation of AATB treated rats. The renal tissues were free from any degeneration of Bowman's space, glomeruli, proximal and distal tubules (Fig. 3).

3.10. LD₅₀ value

The LD₅₀ value for the AATB, calculated using the Acute Oral Toxicity (Guideline 425) Statistical Program (aot425) was above 2000 mg/kg body weight.

4. Discussion

Traditional plant-based therapeutic agents have been getting overwhelming attention and popularity because of the presumption that they are safe, nontoxic and effective in providing the health benefits (Manaharan et al., 2014; Wang et al., 2011; Vaghasiya et al., 2011). The majority of traditional medicaments simply based on its folkloric use that has been perpetuated through generations. Even the most effective traditional herbals may be used as therapeutics only if its toxicity parameters are well assessed (Chanda et al., 2015). In spite of the health benefits they have, many plants possess serious adverse effects too. Many herbal therapeutics are reported to exert severe or even fatal side effects by their improper usage (Tai et al., 1992; Kao et al., 1992). Thorough scientific toxicological evaluation hence forms a prerequisite before a drug is used for its medicinal effects.

Terminalia bellirica (Gaertn.) Roxb is a traditional medicinal plant used for diarrhea, dyspepsia, biliousness, cough, bronchitis and upper respiratory tract infections, tropical pulmonary eosinophilia and allergic eruptions in the traditional systems of medicine (Khare, 2007). Various parts of the plant have been used as a medicinal agent and there are several reports regarding the medicinal value of the plant (Desai et al., 2014; Makihara et al., 2012; Dhingra and Valecha, 2007; Rashed et al., 2014). Even though the plant possesses elite place in the traditional systems of medicine, toxicological evaluation regarding the use of various extracts as well as different parts of the plant is lacking. The literature suggested that only few toxicity studies have been conducted in this plant (Thanabhorn et al., 2006). The present study was hence carried out to evaluate the acute toxicity effect of aqueous acetone extract of *Terminalia bellirica* (Gaertn.) Roxb. fruits.

The acute toxicity study of AATB did not show any signs of toxicity even at the highest dose of 2000 mg/kg body weight. The animals were found to be normal in appearance with no changes in their skin, fur, eyes and mucous membranes. They showed neither behavioral changes such as lethargy, altered sleep patterns nor any observable signs of toxicity such as convulsion, tremors, diarrhea, coma or mortality. The weight gain of AATB treated rats during the study period was normal when compared to the un-treated rats. The loss of weight or abnormal weight gain may be regarded as an indication of toxicity (Teo et al., 2002). Proper food and water intake ensure proper safety evaluation of therapeutic agents in animal models (Kumar and Lalitha, 2013). The normal food and water consumption observed during the study period indicated normal metabolism and health status of treated animals.

In our study, no significant change was observed in the absolute weight of internal organs such as liver, heart, kidneys, lungs and pancreas of AATB treated rats. According to Demma et al., relative organ weight is a more specific parameter than absolute weight in toxicity evaluation in animal models (Demma et al., 2007). Administration of the extract, even at its highest concentration did not show any signs of toxicity with regard to relative organ weight. The hematological parameters indicate physiological as well as the pathological status of animals and are highly sensitive indicators of

Table 2
Relative organ weights of rats receiving AATB extract (Dose in mg/kg body weight) for 14 days.

Experimental groups	Liver	Kidney	Heart	Lungs	Pancreas
Normal	3.66 $\pm$ 0.06	0.95 $\pm$ 0.11	0.62 $\pm$ 0.02	0.86 $\pm$ 0.08	0.62 $\pm$ 0.02
AATB- 55	3.54 $\pm$ 0.05	0.98 $\pm$ 0.02	0.59 $\pm$ 0.03	0.93 $\pm$ 0.01	0.64 $\pm$ 0.03
AATB- 175	3.63 $\pm$ 0.11	1.00 $\pm$ 0.13	0.60 $\pm$ 0.05	0.90 $\pm$ 0.04	0.63 $\pm$ 0.03
AATB-550	3.51 $\pm$ 0.17	0.95 $\pm$ 0.05	0.61 $\pm$ 0.03	0.88 $\pm$ 0.04	0.61 $\pm$ 0.02
AATB-2000	3.48 $\pm$ 0.09	0.96 $\pm$ 0.09	0.58 $\pm$ 0.02	0.87 $\pm$ 0.05	0.59 $\pm$ 0.02

Values are expressed as mean $\pm$ standard deviation ($n = 6$ for each group).

Table 3

Hematological parameters of rats receiving AATB extract (Dose in mg/kg body weight) for 14 days.

Experimental groups	ESR (mm/hr)	Hb (g/dL)	Leukocytes ($\times 10^3/\mu\text{L}$)	Platelets ($\times 10^3 \text{ mm}^3$)	Erythrocytes ($\times 10^6/\mu\text{L}$)
Normal	0.025 $\pm$ 0.013	13.63 $\pm$ 1.25	6.10 $\pm$ 0.12	815.00 $\pm$ 9.02	4.20 $\pm$ 0.07
AATB- 55	0.012 $\pm$ 0.005	13.33 $\pm$ 0.50	6.16 $\pm$ 0.14	822.00 $\pm$ 2.94	4.28 $\pm$ 0.07
AATB- 175	0.037 $\pm$ 0.007	12.08 $\pm$ 0.60	5.89 $\pm$ 0.07	806.00 $\pm$ 4.97	4.29 $\pm$ 0.03
AATB-550	0.018 $\pm$ 0.010	14.83 $\pm$ 1.32	6.26 $\pm$ 0.05	804.80 $\pm$ 5.12	4.10 $\pm$ 0.08
AATB-2000	0.034 $\pm$ 0.007	15.04 $\pm$ 0.47	5.96 $\pm$ 0.13	824.30 $\pm$ 4.57	4.07 $\pm$ 0.08

Values are expressed as mean $\pm$ standard deviation (n = 6 for each group).**Table 4**

Effect of AATB extract (Dose in mg/kg body weight) on biochemical parameters in acute oral toxicity study.

Experimental groups	AST (U/L)	ALP (U/L)	ALT (U/L)	GGT (U/L)
Normal	219.00 $\pm$ 2.16	365.40 $\pm$ 1.11	66.25 $\pm$ 2.22	11.05 $\pm$ 0.76
AATB- 55	217.30 $\pm$ 1.71	367.40 $\pm$ 1.38	64.50 $\pm$ 3.11	11.40 $\pm$ 0.85
AATB- 175	215.30 $\pm$ 2.75	363.30 $\pm$ 2.22	63.75 $\pm$ 3.10	11.30 $\pm$ 0.57
AATB-550	219.30 $\pm$ 1.71	366.00 $\pm$ 0.82	66.50 $\pm$ 2.89	10.88 $\pm$ 1.03
AATB-2000	220.50 $\pm$ 2.65	364.30 $\pm$ 1.25	65.25 $\pm$ 2.63	10.58 $\pm$ 0.75

Values are expressed as the mean $\pm$ standard deviation (n = 6 for each group).**Table 5**

Effect of AATB extract (Dose in mg/kg body weight) on biochemical parameters in acute oral toxicity study.

Experimental groups	A/G ratio	Albumin (g/dL)	Globulin (g/dL)	Bilirubin (mg/dL)	LDH (U/L)
Normal	1.24 $\pm$ 0.14	3.25 $\pm$ 0.24	2.65 $\pm$ 0.33	0.78 $\pm$ 0.10	360.80 $\pm$ 2.75
AATB- 55	1.20 $\pm$ 0.15	3.40 $\pm$ 0.29	2.85 $\pm$ 0.10	0.79 $\pm$ 0.13	355.20 $\pm$ 2.79
AATB- 175	1.24 $\pm$ 0.09	3.35 $\pm$ 0.13	2.68 $\pm$ 0.17	0.74 $\pm$ 0.11	364.30 $\pm$ 4.03
AATB-550	1.23 $\pm$ 0.07	3.28 $\pm$ 0.17	2.63 $\pm$ 0.13	0.73 $\pm$ 0.13	358.50 $\pm$ 4.04
AATB-2000	1.33 $\pm$ 0.18	3.30 $\pm$ 0.18	2.55 $\pm$ 0.39	0.75 $\pm$ 0.10	362.70 $\pm$ 5.03

Values are expressed as mean $\pm$ standard deviation (n = 6 for each group).**Table 6**

Effect of AATB extract (Dose in mg/kg body weight) on renal parameters in acute oral toxicity study.

Experimental groups	Urea (mg/dL)	Creatinine (mg/dL)	Uric acid (mg/dL)	Total protein (g/dL)
Normal	25.50 $\pm$ 3.11	0.27 $\pm$ 0.06	1.95 $\pm$ 0.13	5.90 $\pm$ 0.48
AATB- 55	27.75 $\pm$ 2.75	0.22 $\pm$ 0.02	2.04 $\pm$ 0.11	6.25 $\pm$ 0.21
AATB- 175	29.00 $\pm$ 2.16	0.21 $\pm$ 0.01	1.87 $\pm$ 0.10	5.93 $\pm$ 0.33
AATB-550	21.25 $\pm$ 1.71	0.27 $\pm$ 0.03	1.91 $\pm$ 0.08	5.95 $\pm$ 0.30
AATB-2000	25.00 $\pm$ 3.37	0.22 $\pm$ 0.03	2.14 $\pm$ 0.08	5.85 $\pm$ 0.34

Values are expressed as mean $\pm$ standard deviation (n = 6 for each group).**Table 7**

Effect of AATB extract (Dose in mg/kg body weight) on tissue antioxidant status in the acute oral toxicity study.

Experimental groups	SOD (U/mg protein)	Catalase (U/mg protein)	GSH ($\mu\text{g/g}$ of tissue)	MDA (nmol/g tissue)
Normal	20.59 $\pm$ 1.41	10.50 $\pm$ 0.72	14.58 $\pm$ 0.79	0.42 $\pm$ 0.02
AATB- 55	19.80 $\pm$ 1.07	10.69 $\pm$ 0.97	15.12 $\pm$ 0.88	0.41 $\pm$ 0.02
AATB- 175	20.22 $\pm$ 1.67	10.53 $\pm$ 0.72	14.91 $\pm$ 0.76	0.39 $\pm$ 0.04
AATB-550	20.13 $\pm$ 1.54	10.41 $\pm$ 0.64	14.35 $\pm$ 0.96	0.42 $\pm$ 0.04
AATB-2000	19.79 $\pm$ 1.75	10.23 $\pm$ 0.90	14.55 $\pm$ 0.30	0.40 $\pm$ 0.03

Values are expressed as mean $\pm$ standard deviation (n = 6 for each group).

drug-induced toxicity (Adeneye et al., 2006). The hematological parameters in our study remained normal after the AATB treatment, indicating the nontoxic nature of the extract.

Biochemical parameters are crucial indicators of dysfunction or damage of organs (Balogun et al., 2014; Boehm et al., 2007). Treatment with AATB did not show significant variation in the liver and kidney marker profiles. The liver and kidney are most vulnerable organs to drug-induced toxicity as these organs are primarily involved in the detoxification process. Enzymatic as well as nonenzymatic liver biochemical parameters are of great significance in assessing the liver function and cellular integrity (Chikere,

2015). The enzymes ALT, AST are liver leakage enzymes released upon tissue damage and their amount in peripheral blood stream proportionate with the severity and duration the damage (Ramaiah, 2007). LDH and ALP are also leakage enzymes occasionally referred in the literature as hepatocellular markers of toxicity (Ramaiah, 2007; Manaharan et al., 2014; Suryavanshi et al., 2015). Elevated levels of GGT and ALP serve as a specific biomarker of hepatobiliary obstruction and cholestatic induction (Ramaiah, 2007). The nonenzymatic markers of liver toxicity included in the study were serum total protein, albumin, globulin, bilirubin and A/G ratio. The renal toxicity is indicated by the elevated levels of

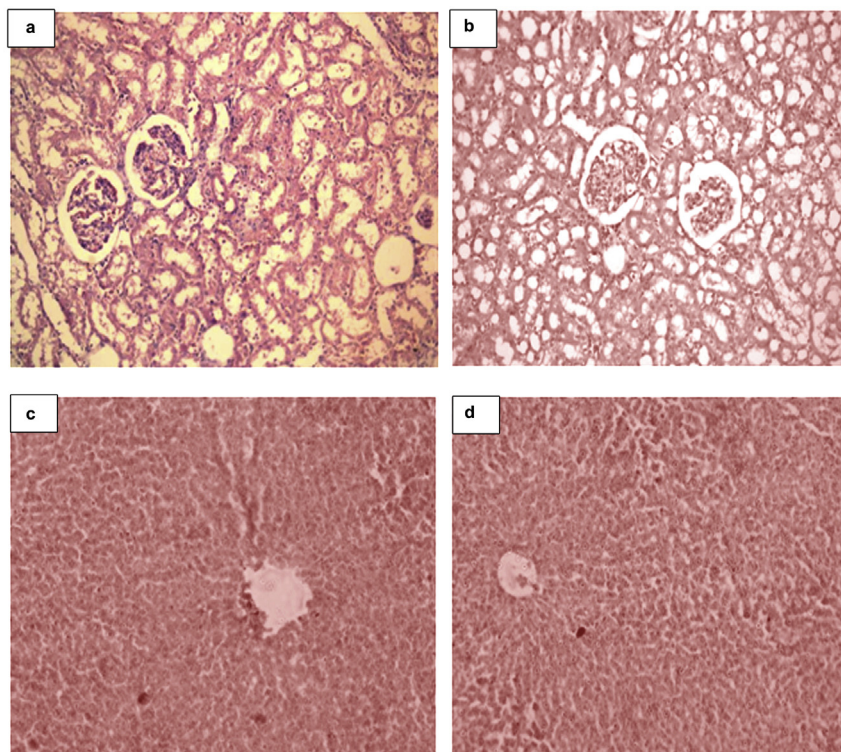


Fig. 3. Histopathology of liver and kidney of control and AATB treated rats at a single dosage of 2000 mg/kg b. wt. in the acute toxicity study for 14 days. The liver showing hepatocytes arranged around central veins with no evidence of necrosis, lesion or any Pathological damage (H&E, 20 ×), and the kidney, showing adequate glomeruli and normal tubules, with no evidence of glomerular damage or lumen casts, (H&E, 20 ×).

creatinine, urea and uric acid in the blood stream due to the poor clearance of these waste products by the impaired kidneys (Wang et al., 2011). In our study, all the biochemical parameters used to assess liver and kidney functions were found to be in the normal range for AATB treated animals in all the four different doses of the extract used. All the values were comparable to the normal untreated animals and hence the results rule out any acute toxicity of the extract.

Macroscopic examination revealed no visible toxicity associated changes in the vital organs of AATB administered rats. Being the primary organs that deal with toxin metabolism, the liver and kidney are very much prone to the pathological changes and hence these organs were selected for detailed histopathological evaluation.

Current research in toxicology is more focused to use female rats for the evaluation safety aspects of plant-based therapeutics in view that female rats are more susceptible to toxic conditions, also, the pathological changes are more pronounced and well expressed (Ping and Darah, 2013). Based on our toxicity study, the LD₅₀ value of AATB should be greater than 2000 mg/kg body weight. As per the OECD guidelines, AATB may be classified under class 5 substance having lethal dose (LD₅₀) higher than 2000 mg/kg body weight.

Terminalia bellirica fruits are widely used to treat several ailments in Indian traditional system of medicine, especially in Ayurveda for centuries but its efficiency is poorly documented through clinical studies. Future studies should focus on placebo controlled, randomized, double-blind clinical trials of the plant based drugs for treatment, outcome, and assessment of adverse herb reactions. This approach will provide figures of risk/benefit profiles that hopefully are positive for at least some treatment modalities of herbal drug (Teschke et al., 2015).

5. Conclusion

The present study revealed that aqueous acetone extract of *Terminalia bellirica* did not produce any acute toxicity in the administered rats. The AATB was found to have the LD₅₀ value higher than 2000 mg/kg body weight in Wistar rats. There was no mortality or adverse effects observed during the study. The biochemical markers all were in the normal range of control animals. Histopathological evaluation also confirmed the nontoxic nature of the extract as its administration showed no remarked changes in the tissue histology. Through our study, it has been established that AATB is nontoxic with an LD₅₀ value greater than 2000 mg/kg body weight. The extract can be developed as a natural therapeutic agent after chronic toxicity evaluation.

Conflicts of interest

Authors would like to disclose that there is no conflicts of interest.

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Transparency document

Transparency document related to this article can be found online at <http://dx.doi.org/10.1016/j.yrtph.2017.04.002>.

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